

equipose

The Mathematics of Posture-Angle Measurement

AI Vision Posture Monitoring for Pediatric Hippotherapy (Cerebral Palsy) • ggmr • June 18, 2026

This note documents exactly how **equipose** turns pose landmarks into the posture angles it reports. Every value is computed in the *two-dimensional image plane* from the pixel coordinates of body landmarks; there is no depth or 3D reconstruction. Two small geometric primitives, a line-tilt and a three-point joint angle, generate all eleven metrics. We give the precise formulas, a worked numerical example, the per-frame to per-session aggregation, and the validity conditions under which the numbers mean what they claim to.

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1 The measurement model

A pose backend (MediaPipe Pose, or MoveNet as a fallback) returns, for one image, a set of named landmarks. Each landmark is a triple

$$\ell_i = (x_i, y_i, c_i), \quad c_i \in [0, 1],$$

where (x_i, y_i) are pixel coordinates and c_i is the backend's *visibility / confidence*. The image origin is the *top-left* corner and the y -axis points *downward*, the usual raster convention. Where geometry only needs position we write a point as $a = (a_x, a_y)$ and drop the confidence.

Two facts that shape everything below.

1. These are *projective* angles measured in one camera's image plane. They approximate true anatomical angles only when the camera is perpendicular to the plane of interest (see §6).
2. The confidence c_i is the lever for occlusion: a metric is computed only when the landmarks it needs are confidently visible, otherwise it is skipped and flagged.

2 Three geometric primitives

Everything reduces to three pure functions of landmark coordinates.

2.1 Line tilt: angle of a two-point line from an axis

For two landmarks a, b we measure the inclination of the line through them. Because a tilt should not depend on which endpoint we list first, we fold the raw angle into $[-90^\circ, +90^\circ]$:

$$\text{wrap}_{90}(\phi) = \begin{cases} \phi - 180^\circ, & \phi > 90^\circ, \\ \phi + 180^\circ, & \phi < -90^\circ, \\ \phi, & \text{otherwise.} \end{cases}$$

Tilt from horizontal ($0^\circ = \text{level}$) and **tilt from vertical** ($0^\circ = \text{plumb}$):

$$\begin{aligned} \phi_H(a, b) &= \text{wrap}_{90}(\text{atan2}(b_y - a_y, b_x - a_x)), \\ \phi_V(a, b) &= \text{wrap}_{90}(\text{atan2}(b_x - a_x, b_y - a_y)). \end{aligned}$$

The swapped argument order in ϕ_V measures from the vertical axis instead of the horizontal. The sign encodes direction (because y grows downward, a positive ϕ_H slopes down to the right).

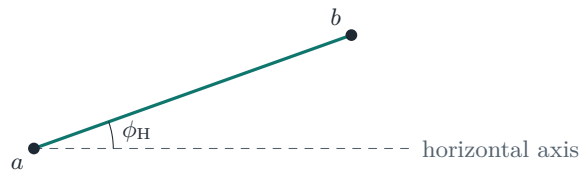


Figure 1. Line tilt $\phi_H(a, b)$: the angle of the landmark line above the horizontal.

2.2 Joint angle: the angle at a middle joint

For three landmarks a, m, b the angle *at* m is obtained from the dot product of the two limb-segment vectors $u = a - m$ and $v = b - m$:

$$\theta(a, m, b) = \arccos\left(\frac{(a - m) \cdot (b - m)}{\|a - m\| \|b - m\|}\right) \in [0^\circ, 180^\circ].$$

A straight limb gives $\theta = 180^\circ$; a right angle gives 90° . The cosine argument is clamped to $[-1, 1]$ for numerical safety, and a degenerate (zero-length) segment returns 180° by convention.

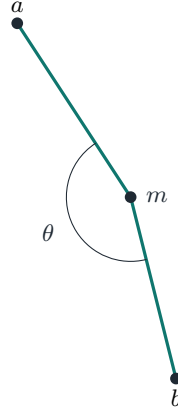


Figure 2. Joint angle $\theta(a, m, b)$ at the vertex m , between segments $m \rightarrow a$ and $m \rightarrow b$.

2.3 Perpendicular distance: a point off an axis

The signed-free distance from a point p to the infinite line through a and b (used for midline deviation):

$$d(p; a, b) = \frac{|(b_y - a_y)p_x - (b_x - a_x)p_y + b_x a_y - b_y a_x|}{\sqrt{(b_y - a_y)^2 + (b_x - a_x)^2}}.$$

Two convenience helpers complete the toolkit: the midpoint $\text{mid}(a, b) = \frac{1}{2}(a + b)$ and the segment length $\|a - b\|$ (used to normalize pixel distances by shoulder width).

3 The eleven posture metrics

Each metric is one primitive applied to specific landmarks. Metrics split by camera view, and by *reliability*: **primary** metrics use head / shoulder / trunk landmarks that stay visible above the saddle; **best-effort** metrics use pelvis / hip / knee landmarks that the horse and side-walkers frequently occlude.

3.1 Front view (coronal plane)

Metric	Tier	Definition
head_tilt	primary	$\phi_H(\ell_{L \text{ ear}}, \ell_{R \text{ ear}})$ (falls back to eyes)
shoulder_tilt	primary	$\phi_H(\ell_{L \text{ sh}}, \ell_{R \text{ sh}})$
pelvic_obliquity	best-effort	$\phi_H(\ell_{L \text{ hip}}, \ell_{R \text{ hip}})$
trunk_lateral_lean	primary	$\phi_V(\text{mid}(\ell_{L \text{ sh}}, \ell_{R \text{ sh}}), \text{mid}(\ell_{L \text{ hip}}, \ell_{R \text{ hip}}))$
midline_deviation	primary	$\frac{d(\ell_{\text{nose}}; \text{mid}_{\text{sh}}, \text{mid}_{\text{hip}})}{\ \ell_{L \text{ sh}} - \ell_{R \text{ sh}}\ }$ (fraction of shoulder width)
symmetry_score	primary	composite, see §3.3

3.2 Side view (sagittal plane)

The camera-facing side $s \in \{L, R\}$ is chosen automatically as the side with higher mean landmark confidence over a representative joint set $J = \{\text{shoulder, hip, ear, knee, ankle}\}$:

$$s = \arg \max_{s \in \{L, R\}} \frac{1}{|J_s|} \sum_{j \in J_s} c_j.$$

Metric	Tier	Definition (side s)
forward_trunk_lean	primary	$\phi_V(\ell_{s, \text{sh}}, \ell_{s, \text{hip}})$ (falls back to ear→shoulder)
neck_forward_angle	primary	$\theta(\ell_{s, \text{ear}}, \ell_{s, \text{sh}}, \ell_{s, \text{hip}})$
hip_flexion	best-effort	$\theta(\ell_{s, \text{sh}}, \ell_{s, \text{hip}}, \ell_{s, \text{knee}})$
knee_angle	best-effort	$\theta(\ell_{s, \text{hip}}, \ell_{s, \text{knee}}, \ell_{s, \text{ankle}})$
sitting_alignment_index	primary	composite, see §3.3

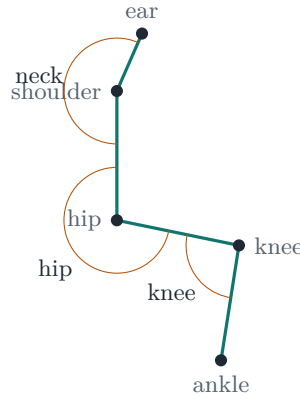


Figure 3. Sagittal landmark chain. *neck_forward_angle*, *hip_flexion*, and *knee_angle* are the joint angles at shoulder, hip, and knee; *forward_trunk_lean* is the tilt of the shoulder→hip segment from vertical.

3.3 Composite indices

Two metrics summarize several others into a single 0–100 score (higher is better). They are deliberately simple weighted penalties; the scaling constants are provisional and pinned for calibration

at the pilot.

$$\begin{aligned}\text{symmetry_score} &= \max\left(0, 100 - (|\text{head_tilt}| + |\text{shoulder_tilt}| + 100|\text{midline_deviation}|)\right), \\ \text{sitting_alignment_index} &= \max\left(0, 100 - (|\text{forward_trunk_lean}| + |180^\circ - \text{neck_forward_angle}|)\right).\end{aligned}$$

4 Worked example

Take three landmarks from a side view (pixels): ear $a = (90, 80)$, shoulder $m = (100, 100)$, hip $b = (100, 200)$. Then $\text{neck_forward_angle} = \theta(a, m, b)$:

$$\begin{aligned}u &= a - m = (-10, -20), & v &= b - m = (0, 100), \\ u \cdot v &= (-10)(0) + (-20)(100) = -2000, & \|u\| &= \sqrt{500}, \quad \|v\| = 100, \\ \cos \theta &= \frac{-2000}{100\sqrt{500}} \approx -0.8944, & \theta &= \arccos(-0.8944) \approx 153.4^\circ.\end{aligned}$$

A forward-flexed neck (head dropping toward the chest) would pull this angle *down* from 180° ; here 153° reflects a mild forward lean. The unit tests pin exactly these reference values.

5 From a measurement to a reported value

5.1 Confidence gating (occlusion handling)

A metric is evaluated only when *every* landmark it requires clears the visibility threshold τ (default $\tau = 0.3$); otherwise its value for that frame is undefined ($\emptyset$) and flagged. The reported per-metric confidence is the minimum confidence over the landmarks used:

$$c_{\text{metric}} = \min_{j \in \text{used}} c_j, \quad \text{value defined} \iff c_{\text{metric}} \geq \tau.$$

This is why the lower-body, best-effort metrics legitimately drop out on a pony: the saddle and side-walkers push hip / knee / ankle confidence below τ .

5.2 Temporal smoothing and aggregation (video)

For a video, each metric yields a time series. Before aggregation the series is passed through a *One-Euro* filter to suppress per-frame jitter while staying responsive; short gaps of missing frames are interpolated and long gaps are preserved as $\emptyset$. Over the valid frames $\mathcal{V} = \{t : v_t \neq \emptyset\}$ with acceptable range $[\ell, h]$:

$$\begin{aligned}\bar{v} &= \frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} v_t, & \sigma &= \sqrt{\frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} (v_t - \bar{v})^2}, \\ p_{\text{in}} &= \frac{1}{|\mathcal{V}|} \sum_{t \in \mathcal{V}} \mathbf{1}[\ell \leq v_t \leq h], & \delta_{\text{max}} &= \max_{t \in \mathcal{V}} \max(0, \ell - v_t, v_t - h).\end{aligned}$$

Here p_{in} is the fraction of the session spent inside the acceptable range, and δ_{max} is the worst excursion outside it.

5.3 The single-photo case

A photo is one frame, so $|\mathcal{V}| = 1$: there is no smoothing and no spread. The aggregates collapse to

$$\bar{v} = v, \quad \sigma = 0, \quad p_{\text{in}} \in \{0, 1\},$$

i.e. the reported value *is* the single measurement and “% in range” becomes a binary in/out flag. equipose therefore labels photo output a *snapshot*, not a trend.

6 Validity and limitations

These are 2D projective approximations, not goniometer readings. A single camera measures angles in its own image plane. The numbers approximate the true 3D anatomical angles only when the optical axis is perpendicular to the relevant plane: the *front* camera for coronal tilts, the *side* camera for sagittal flexion. Any out-of-plane body rotation introduces error that one view cannot recover.

Three practical consequences:

- **Camera setup is part of the measurement.** Perpendicular optical axis, camera at the child’s trunk height, full body in frame. These are prerequisites, documented in `SETUP_PROTOCOL.md`.
- **Thresholds are placeholders.** The acceptable ranges $[\ell, h]$ behind every in/out flag, and the composite scaling constants, were not derived from clinical data. They are tunable defaults to be calibrated against clinician judgement at the pilot.
- **Lower-body metrics are best-effort by design.** On a pony the pelvis and legs are the most occluded and the most clinically interesting; equipose reports them when confident and abstains when not, rather than fabricating a number.

equipose is a research and screening aid, not a diagnostic device. Clinical decisions remain with the therapist and physician. Source of truth for these formulas: `geometry.py`, `angles_front.py`, `angles_side.py`, and `aggregate.py`.